

SENG1050

Lab 6 (Linux)

In this lab, you will gain experience in configuring the Apache web server and create custom behaviour using PHP.

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Task 1 - Install XAMPP

It is recommended to install XAMPP on your personal computer. The instructions are as follows:

1. Download XAMPP from [here](#).
2. Launch the Terminal application.
Enter the commands:

```
cd Downloads
chmod 755 xampp-linux-x64-8.2.12-0-installer.run
sudo ./xampp-linux-x64-8.2.12-0-installer.run
```

3. You will be presented with the XAMPP installation wizard.
Follow the prompts to complete installation. **Keep all installation settings and directory to default.**
4. Launch the Terminal application.
Enter the commands:

```
sudo chown -R <username>:<username> /opt/lampp
sudo chmod -R 755 /opt/lampp
```

where `<username>` is the name of your account.
This changes folder ownership to make development easier.

5. In a new Terminal window, enter the command:

```
sudo /opt/lampp/manager-linux-x64.run
```

This launches `manager-linux-x64.run`, which is the XAMPP control panel.

6. In the XAMPP Control Panel, navigate to the 'Manage Servers' tab.
Start the 'Apache Web Server' module.
7. In your browser, enter `localhost` in the URL bar. Set up is done if you see the XAMPP Homepage.

In case you can't or do not prefer to install XAMPP on your personal computer, you may use the provided virtual machine. Instructions to access the virtual machine are as follows (**skip to Task 2 if you have XAMPP installed**):

1. Start your virtual machine from [Azure Labs](#) (use your regular student email to login) and wait until the VM has started.
If you cannot see the SENG1050 VM in the list, check your email for an invite link and register.
2. Click the small computer icon to download a connection link (SENG1050.rdp)
3. Before connecting to the virtual machine, you have to download and install [Remmina](#) as your remote desktop client.
4. Double-click on the rdp file and login to the VM using the password 'K43NJR9nHEsfbJ' (without the quotes).
You may have to set the Username as AzLabImageAdmin (without '~\') and Domain as WORKGROUP.

Ignore any certificate error message and click 'Yes' to proceed.

5. If you are using the VM, you have to follow the Windows version of our lab manual to complete the XAMPP installation and all lab tasks.

Task 2 - Configure Apache Web Server

In this task, you are going to learn how to configure the Apache web server to host a website and serve content securely. You do not need to be familiar with every possible configuration setting in Apache. The main aim of this task is for you to become familiar with how to host a website, and how to adjust commonly-used security settings. In case you are stuck or are curious about the reasoning behind these settings, you may refer to the supplementary video on Canvas (under the manual).

1. Start the XAMPP Control Panel (`manager-linux-x64.run`).
2. In the XAMPP Control Panel, start the Apache module.
3. Navigate to the default document root of XAMPP, which is `/opt/lampp/htdocs`.
In this folder, create a new directory for your website. Name the directory `seng1050` (Note that you can change this to any name).
4. Create a copy of your `template.html` file from Lab 2, and place it in the `seng1050` directory.
Rename the copy to `index.html`.
5. Navigate to `/opt/lampp/etc`.
Open the `httpd.conf` file.
Uncomment the line (Remove '#'): `Include etc/extra/httpd-vhosts.conf`
6. Navigate to `/opt/lampp/etc/extra`.
Open the `httpd-vhosts.conf` file.
Include the following lines at the end of the file:

```
<VirtualHost *:80>
    DocumentRoot "/opt/lampp/htdocs/seng1050"
    ServerName seng1050.com
    <Directory "/opt/lampp/htdocs/seng1050">
        Options Indexes
    </Directory>
</VirtualHost>
```

7. Launch the Terminal application.
Type the command `sudo nano /etc/hosts` to edit the `hosts` file in `/etc`.
Add the following line at the end of the file:

```
127.0.0.1 seng1050.com
```

Save the file by pressing `^O`.

Note that you can change `seng1050.com` to any preferred website address as long as it matches the `ServerName` in Step 6.

8. Restart Apache in the XAMPP control panel.
9. Open the browser and enter `http://seng1050.com` in the URL bar.
You should see your `index.html` file in the browser.

Now with the virtual host up and running, we will look at some commonly used Apache security configuration settings.

1. First, we will remove the OS and version information Apache sends in every HTTP response.
In the browser, open the Inspector and select the Network tab.
Refresh the page and look at the headers of the response.
Notice the field 'Server' that exposes the version of Apache and our OS? We want to remove as much information from this as possible.
2. Navigate to `/opt/lampp/etc`.
Open the `httpd.conf` file.
Include the following lines at the end of the file:

3. Save the file and restart Apache in the XAMPP control panel.
4. Refresh the page in the browser and view the response headers in the inspector again.
If you have set the configuration correctly, only the name of the web server, 'Apache' should be shown in the 'Server' field.
5. Second, we will remove information about our website directory.
Rename the `index.html` file in the `seng1050` folder to `index_temp.html`.
Refresh the page and you will notice that without finding the default document, Apache shows a listing of all files in the directory. This can expose information we want to keep secret. Let's fix this now!
6. Navigate to `/opt/lampp/etc/extra`.
Open the `httpd-vhosts.conf` file.
Change the line `Options Indexes` to `Options -Indexes`
7. Save the file and restart Apache in the XAMPP control panel.
8. Refresh the page in the browser.
You should now see a 403 Forbidden error message instead of a full directory listing.

If you're interested in more information on securing and configuring Apache, details can be found [here](#).

Task 3 - Basic PHP

In this task, you are going to learn how to use PHP to create and send HTML documents to the client.

1. Create a new file in your website called `index.php`. Copy the following code into the file:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Hello World</title>
  </head>
  <body>
    <?php
      echo "<p>Hello World</p>";
    ?>
  </body>
</html>
```

2. Now, open your website in the browser (e.g., `http://seng1050.com`).
As `index.php` is one of the default documents in Apache, you should see the file load, with "Hello World" displayed on the page.
3. Next, we will edit the `index.php` file to generate a countdown from 10 to 0 using a while loop.
4. Replace the echo statement in the PHP script with the following:

```
$counter = 10;
while ($counter >= 0) {
  echo "<p>" . $counter-- . "</p>";
}
```

5. Save the file and reload the page in your browser. You should see a countdown from 10 to 0.
6. Now, we will edit the file to display only even numbers in the countdown. In the PHP script, add the following code just before the echo statement within the 'while' loop:

```
if ($counter % 2 != 0) {
  $counter--;
  continue;
}
```

7. Save the file and reload the page in your browser. You should see a countdown from 10 to 0 with only the even numbers displayed.

8. Now, it's time for you to try some PHP tasks on your own. Edit your PHP file to perform the following tasks:
 - Declare the variable `$the_string = "The cat sat on the mat";`. Calculate the length of this string, as well as the number of times the letter 't' appears in it. (Hint: Use `strlen` and `substr_count`.)
 - Display the basic times-tables for 1x1 through to 12x12.
 - Display all the leap years from 1900 to 2022. (Hint: Leap years are the years that are either: evenly divisible by 4 AND not evenly divisible by 100; OR: evenly divisible by 4 AND 400.)

If you run into problems with your PHP code, try outputting the value of variables with echo statements, or checking for syntax errors with a [PHP tester](#).

Task 4 - PHP With Form Data

PHP becomes very powerful when used with dynamic data collected from the user via a form. In this task, we will learn to use form data with PHP.

1. Create a new file in your website called `form.html`. Copy the following code into the file:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>My Form</title>
  </head>
  <body>
    <form method="POST" action="index.php">
      <label for="name">Name</label>
      <input type="text" name="fname" id="name" /><br><br>
      <input type="submit">
    </form>
  </body>
</html>
```

2. Remember, the value of the 'name' attribute is what we will use to get the value of the input in our PHP script.
 3. Now, edit your `index.php` file again to get the value of the text box, and display it back to the client. (Hint: Use the super-global array `$_POST["fname"]` to retrieve the value of the input).
 4. Edit your `index.php` file again to perform the following tasks:
 - Display the length of the form input. (Hint: Use `strlen`.)
 - Display the number of times the letter 'e' appears in the input. (Hint: Use `substr_count`.)
 - Display the form input in all upper-case letters. (Hint: Use `strtoupper`.)
 5. Now, in `form.html`, add an additional input `<input type="date" name="dob"/>` which will be used to collect the user's date of birth.
 6. Back in your `index.php` file, use the value of the new date input to perform the following tasks:
 - Display the date of birth in year-month-day format.
 - Calculate and display the age of the user based on difference in years. (Hint: Retrieve the current date using `$today = date('Y-m-d');` and calculate the difference using `date_diff(date_create($dob), date_create($today)) -> y;`)
 - If the user is under the age of 18, output the string: "Sorry, you need to be over the age of 18 to make this purchase".
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Task 5 - Advanced PHP tasks

If you have extra time at the end, try out the following tasks. These are much more challenging, you may need to ask your demonstrator for assistance.

1. Edit your PHP script to output the Fibonacci sequence from 1 up to a number entered by the user via the form (you'll need add an input of type "number" to the form).
The Fibonacci sequence sums the previous two numbers to get the next number in the sequence, e.g: 1, 1, 2, 3, 5, 8, 13, 21, 34, ...
2. Add four required number inputs to the form.
Edit your PHP script to display the second-largest number entered.
(Hint: You may need `array` and `sort`.)
3. Edit your PHP script to operate as a simple calculator.
Take two number inputs from the form, and use the value of the form's text input to determine which operation to perform ('+' for addition, '-' for subtraction, '*' for multiplication, and '/' for division).
Display the result.
4. Edit your PHP script to take a text input as a string.
Display the string with reversed case (e.g., 'Hi There' should be output as 'hI tHERE').
(Hint: You may need `ctype_lower`, `strtolower`, and `strtoupper`.)
5. Edit your PHP script to take the four number inputs from Step 2.
Output these numbers in ascending order.
6. Edit your PHP to take two text inputs from the form.
Output the longest common ending between both inputs (e.g., given the two strings "action" and "addition", the longest common ending would be "tion").